Publications

Katrín Björnsdóttir Google Scholar profile

Peer-reviewed articles

2026

Verdonen M, Barrio IC, Barbero-Palacios L, … **Björnsdóttir K**, … Myers-Smith IH (28 authors). **Borealization of tundra ecosystems with climate and land-use change.** *Environmental Research Letters 21(1): 013002.* doi:10.1088/1748-9326/ae2e18

2025

Barrio IC, Vuorinen KEM, Barbero-Palacios L, … **Björnsdóttir K**, … Kamenova S (79 authors). **Emerging priorities in terrestrial herbivory research in the Arctic.** *Arctic Science 11.* doi:10.1139/as-2024-0080

Schwieger S, Dorrepaal E, Petit Bon M, … **Björnsdóttir K**, … Sarneel JM (42 authors). **Environmental conditions modulate warming effects on plant litter decomposition globally.** *Ecology Letters 28(1): e70026.* doi:10.1111/ele.70026

Eidesen PB, Brysting AK, Hagen KR, … **Björnsdóttir K**, … Volden IK (26 authors). **Ecological and evolutionary consequences of ploidy-driven trait variation: insights from *Saxifraga oppositifolia* L.** *Arctic Science 11.* doi:10.1139/as-2024-0020

2024

Sarneel JM, Hefting MM, Sandén T, … **Björnsdóttir K**, … Keuskamp JA (100 authors). **Reading tea leaves worldwide: decoupled drivers of initial litter decomposition mass-loss rate and stabilization.** *Ecology Letters 27(5): e14415.* doi:10.1111/ele.14415

Barbero-Palacios L, Barrio IC, García Criado M, … **Björnsdóttir K**, … Soininen EM (36 authors). **Herbivore diversity effects on Arctic tundra ecosystems: a systematic review.** *Environmental Evidence 13: 6.* doi:10.1186/s13750-024-00330-9

2023

Vandvik V, Halbritter AH, Althuizen IHJ, … **Björnsdóttir K**, … Enquist BJ (35 authors). **Plant traits and associated data from a warming experiment, a seabird colony, and along elevation in Svalbard.** *Scientific Data 10: 578.* doi:10.1038/s41597-023-02467-7

Jónsdóttir IS, Halbritter AH, Christiansen CT, Althuizen IHJ, Haugum SV, Henn JJ, **Björnsdóttir K**, Maitner BS, Malhi Y, Michaletz ST, Roos RE, Klanderud K, Lee H, Enquist BJ, Vandvik V. **Intraspecific trait variability is a key feature underlying high Arctic plant community resistance to climate warming.** *Ecological Monographs 93(1): e1555.* doi:10.1002/ecm.1555

2022

Lembrechts JJ, van den Hoogen J, Aalto J, … **Björnsdóttir K**, … Lenoir J. **Global maps of soil temperature.** *Global Change Biology 28(9): 3110–3144.* doi:10.1111/gcb.16060

2021 Soininen EM, Barrio IC, Bjørkås R, **Björnsdóttir K**, … Speed JDM (44 authors). **Location of studies and evidence of effects of herbivory on Arctic vegetation: a systematic map.** *Environmental Evidence 10: 25.* doi:10.1186/s13750-021-00240-0

Björnsdóttir K, Barrio IC, Jónsdóttir IS. **Long-term warming manipulations reveal complex decomposition responses across different tundra vegetation types.** *Arctic Science 8(3): 979–991.* doi:10.1139/as-2020-0046

2019 Mulloy TA, Barrio IC, **Björnsdóttir K**, Jónsdóttir IS, Hik DS. **Fertilisers mediate the short-term effects of sheep grazing in the Icelandic highlands.** *Icelandic Agricultural Sciences 32: 75–85.* doi:10.16886/ias.2019.07

Manuscripts

2026 **Björnsdóttir K**, Myers-Smith IH, Thomas H, Pacheco-Riaño LC, Elmendorf SC, Wårlind D, Miller PA, Petit Bon M, Björk RG, Cornelissen JHC, Lang SI, Jónsdóttir IS, Vandvik V, … Bjorkman A. **Plant community litter decomposability increases with warming across the tundra.** *Manuscript ready for submission.*

Björnsdóttir K, Gya R, Bjorkman A. **Temperature sensitivity of tundra plant functional traits to experimental warming.** *Manuscript in preparation.*

2025 Borderieux J, Thomas H, Myers-Smith IH, Høye T, Petit Bon M, Lembrechts J, Walker E, **Björnsdóttir K**, … Sarneel J (40 authors). **Litter quality outweighs climate as a driver of decomposition across the tundra biome.** *In revision, Functional Ecology. Preprint: EcoEvoRxiv.* doi:10.32942/x2q352

Theses

2026 **PhD thesis, University of Gothenburg**

– *Consequences of climate change for ecosystem functioning mediated by plant responses across the tundra.* Department of Biological and Environmental Sciences. hdl.handle.net/2077/93153

2018 **M.Sc. thesis, University of Iceland**

– *Decomposition responses to climate warming and sheep grazing in the high and sub-Arctic.* Advisors: Ingibjörg Svala Jónsdóttir and Isabel C. Barrio. hdl.handle.net/1946/30518

2015 **B.Sc. thesis, University of Iceland**

– *Áhrif vegagerðar í Skaftáreldahrauni á landnám gulvíðis, loðvíðis og krækilyngs.*